

4(a)(i)	<i>any four from:</i> 1 obtain single-stranded DNA ; 2 label DNA with fluorescent, dye / tag / label ; 3 probes on, chip / microarray ; 4 each probe is unique to a particular, SNP / gene / allele ; 5 DNA (added), binds / hybridises, to probes ; 6 wash off excess DNA (after hybridisation) ; 7 (view under) UV light / laser scanner / high resolution digital camera ; 8 fluorescence indicates presence of, SNP / gene ; R gene expressed	4
4(a)(ii)	<i>any two from:</i> 1 large database ; 2 use database to find, SNPs / probes ; 3 fast / accurate / efficient ; 4 <i>ref. to</i> (computer) software / algorithms ; 5 <i>ref. to</i> statistics / statistical analysis ; 6 <i>idea of</i>, $17\,000 \times 500\,000 / 8.5 \times 10^9$, data points ;	2
4(b)(i)	1 and 6 ;	1

Question	Answer	Marks
4(b)(ii)	any two from: 1 all diseases, are autosomal / not associated with X chromosome / not sex-linked ; 2 all diseases are associated with SNP on chromosome 6 ; 3 type 1 and type 2 diabetes SNPs are associated with chromosomes 12 and 16 (but not rheumatoid arthritis) ; 4 type 2 diabetes SNP associated with chromosome 10 (but not type 1 diabetes and rheumatoid arthritis) ; 5 arthritis and type 1 diabetes have higher level of association (with each other) than with type 2 diabetes ; 6 rheumatoid arthritis has fewest number of SNPs or type 1 and type 2 diabetes has more SNPs (than rheumatoid arthritis) ;	2

Question	Answer	Marks
4(c)	any three from: 1 DNA analysis not available for everyone ; 2 lifestyle change ; A e.g. lose weight / stop smoking / take exercise 3 early treatment ; 4 allows people to plan ; e.g. organise care / appoint power of attorney / take out medical insurance / make will / gift property to children / retire early 5 decide whether to have children / ref. (therapeutic) termination ; 6 results may cause, anxiety / stress / panic / depression (if positive) or reduce worry (if negative) ; 7 results may affect ability to get, insurance / credit / jobs ; 8 <i>idea of</i> predictions may not be accurate ;	3

Question	Answer	Marks
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